**(iv) Intercept form**

Equation of a straight line whose $x$ and $y$ intercepts are $a$ and $b$, is $\frac{x}{a}+\frac{y}{b}=1$.

**(v) General form**

Equation of straight line in general form is $ax+by+c=0$ where $a$, $b$ and $c$ are constants and $a$, $b$ are not simultaneously zero.

### 3.2.2 Angle between two straight lines

Let $l_1$ and $l_2$ be two straight lines represented by the equations $l_1: y=m_1x+c_1$ and $l_2: y=m_2x+c_2$ intersecting at $P$.

If $\theta_1$ and $\theta_2$ are two angles made by $l_1$ and $l_2$ with $x$-axis then slope of the lines are $m_1=\tan\theta_1$ and $m_2=\tan\theta_2$.

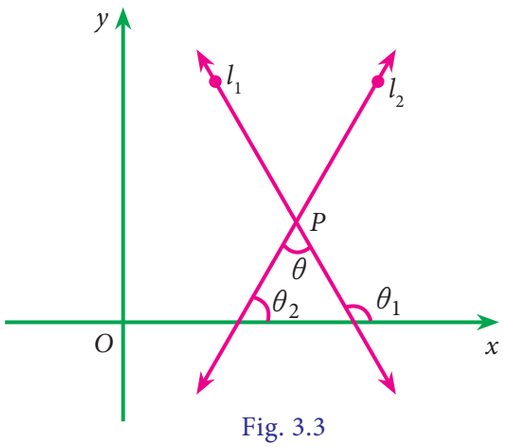



Fig. 3.3

From fig 3.3, if $\theta$ is angle between the lines $l_1$ & $l_2$ then

$$\theta=\theta_1-\theta_2$$

$$\therefore \tan\theta=\left|\tan(\theta_1-\theta_2)\right|$$

$$=\left|\frac{\tan\theta_1-\tan\theta_2}{1+\tan\theta_1\tan\theta_2}\right|$$

$$\tan\theta=\left|\frac{m_1-m_2}{1+m_1m_2}\right|$$

$$\theta=\tan^{-1}\left|\frac{m_1-m_2}{1+m_1m_2}\right|$$

**NOTE**

(i) If $\frac{m_1-m_2}{1+m_1m_2}$ is positive, then $\theta$, the angle between $l_1$ and $l_2$ is acute and if it is negative, then, $\theta$ is the obtuse.

(ii) We know that two straight lines are parallel if and only if their slopes are equal.

(iii) We know that two lines are perpendicular if and only if the product of their slopes is –1. (Here the slopes $m_1$ and $m_2$ are finite.)

**DO YOU KNOW?**

The straight lines $x$-axis and $y$-axis are perpendicular to each other. But, the condition $m_1m_2=-1$ is not true because the slope of the $x$-axis is zero and the slope of the $y$-axis is not defined.

**Example 3.4**

Find the acute angle between the lines $2x-y+3=0$ and $x+y+2=0$

***Solution***

Let $m_1$ and $m_2$ be the slopes of $2x-y+3=0$ and $x+y+2=0$

Now $m_1=2$, $m_2=-1$

Let $\theta$ be the angle between the given lines

$$\tan\theta=\left|\frac{m_1-m_2}{1+m_1m_2}\right|$$

$$=\left|\frac{2+1}{1+2(-1)}\right|=3$$

$$\theta=\tan^{-1}(3)$$



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